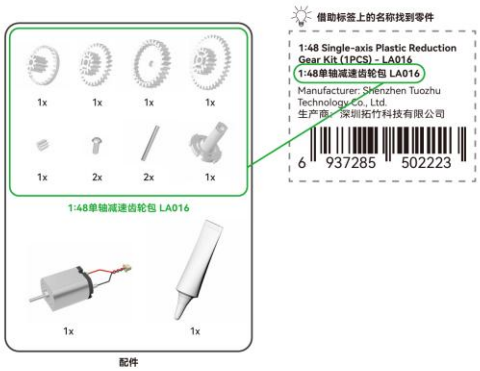
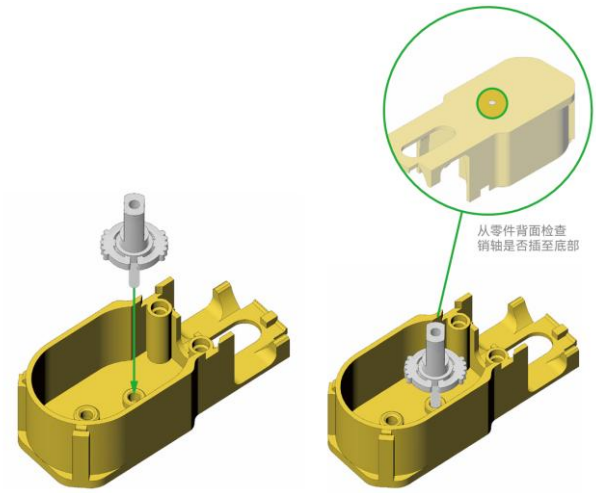
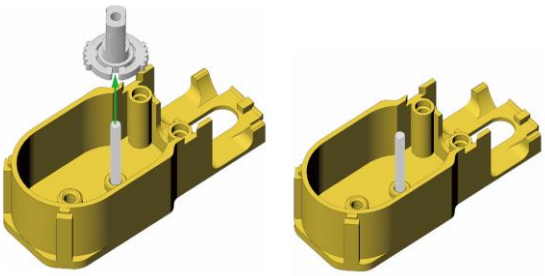
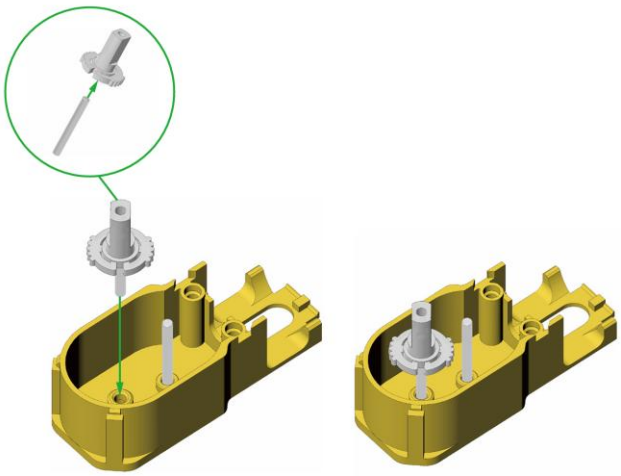
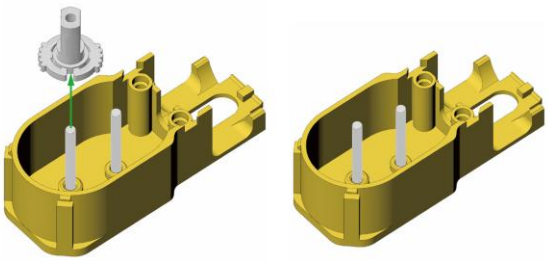
1、安装 1：48 减速箱（英文请查看官方叉车的装配文档：https://makerworld.com.cn/zh/models/1207210-cyberbrick-guan-fang-cha-che?from=search#profileId-1282676）

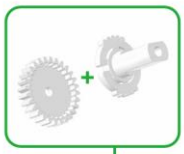


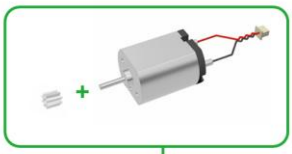


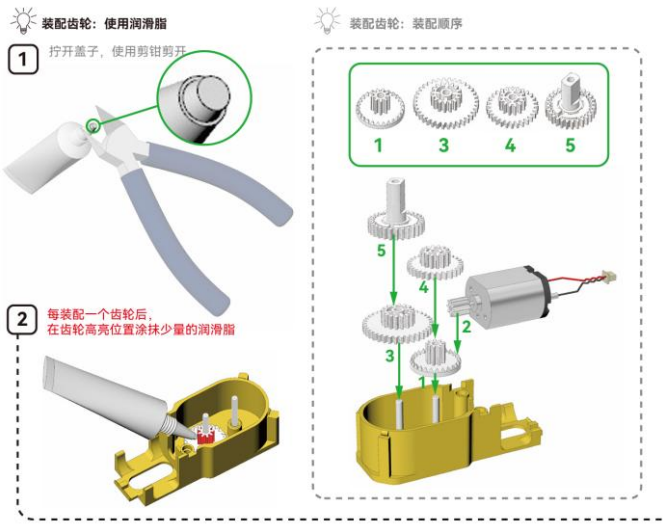


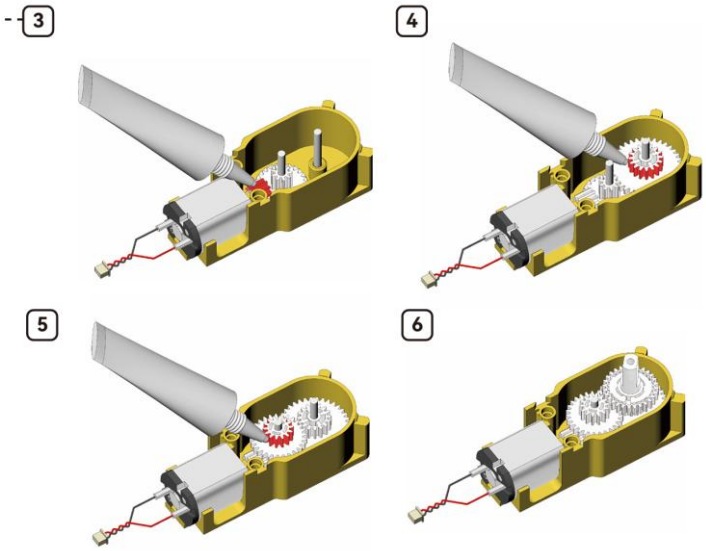


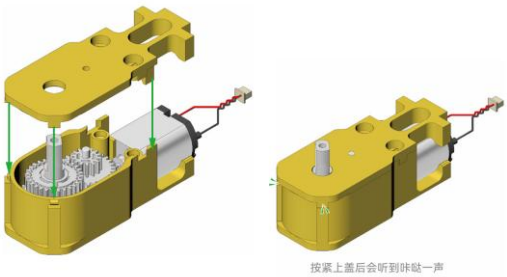


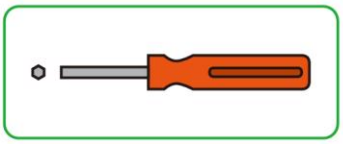






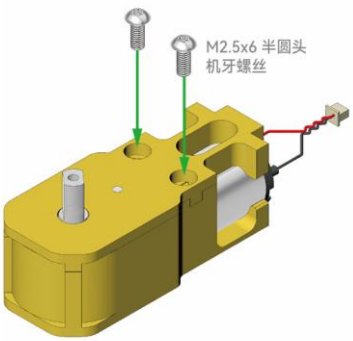




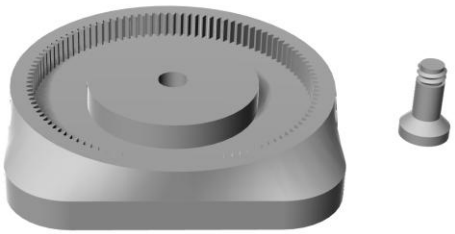


请使用H1.5内六角螺丝刀

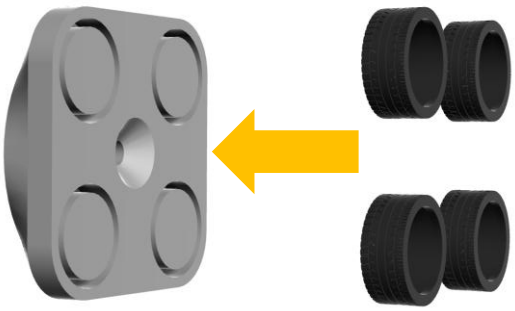
不旋入螺丝也可以，只要按紧上盖即可



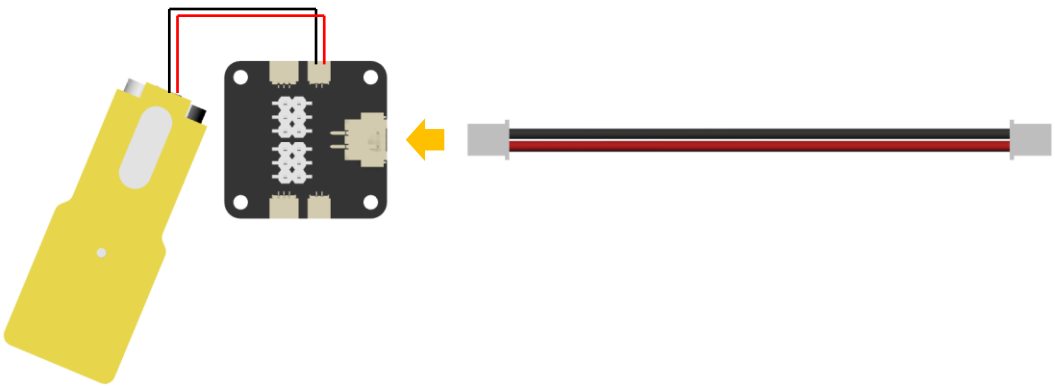
2、选择你需要的底座，上部结构容易拆卸，方便开关电源和拿起把玩的底座（左图）；不易拆卸，一体性较强，能将上部结构固定相连的底座（右图）【后续安装示例将使用右图底座】。



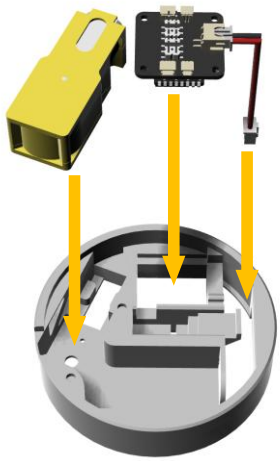
3、将4个轮胎，内部有1圈凸起的部分朝外，嵌入底座底面的孔洞中。安装好后放在一边备用。

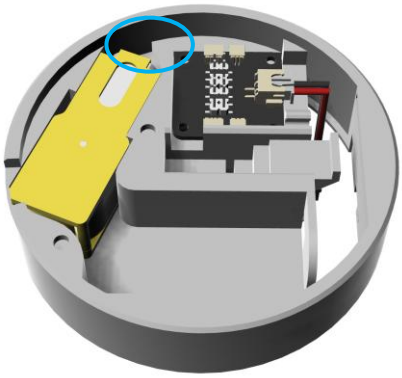


4、将减速电机电源和开关包装中的双头电源线插入接收板对应孔位。

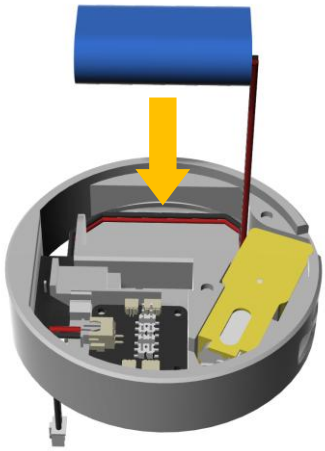


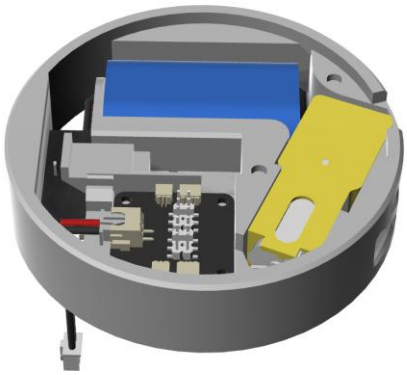
5、将减速电机、接收板和电源线放入旋转底盘的对应孔位，电源线从右侧空洞中穿进，并将过长的电源线折叠放置在蓝色圆圈内。



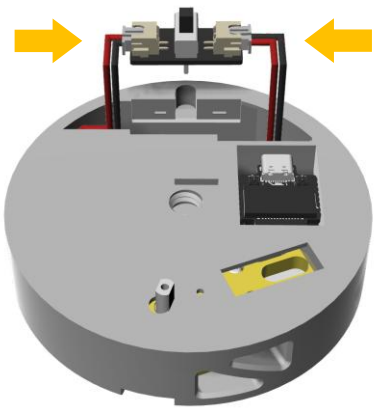


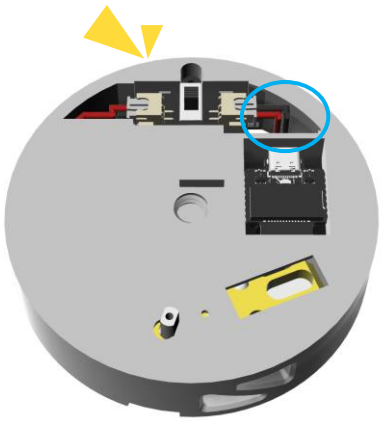
6、把电池的线塞入电池仓内壁的凹陷内，线头向下穿进左侧的空洞内，并且同时将电池放入电池仓



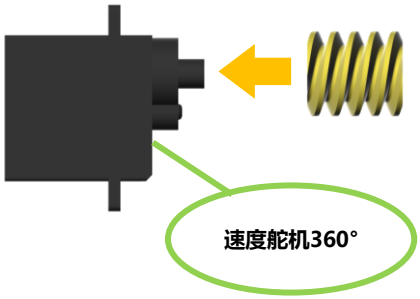


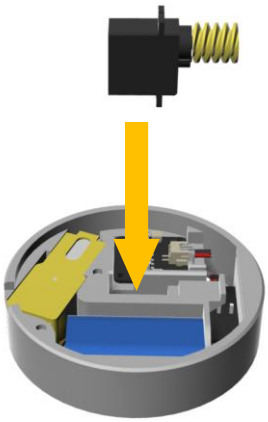
7、翻到旋转底盘背面，将接收器电源线和电池线分别插入开关两侧的接口，并将开关按入底面的槽位内，听到“咔哒”一声表明安装到位。并将接收板过长的电源线折叠，塞入蓝色圆圈内的空位。如果此时开关是开启的状态，请将其关闭。



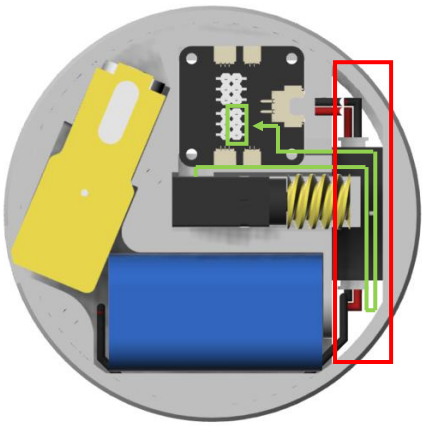


8、将蜗杆插入速度舵机的转轴上，并尽可能的按压。之后将舵机和蜗杆组合体插入旋转底盘对应的槽位内。

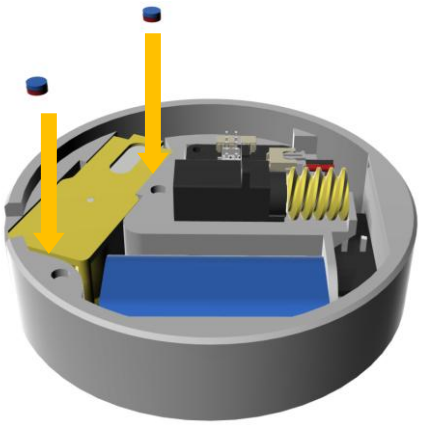


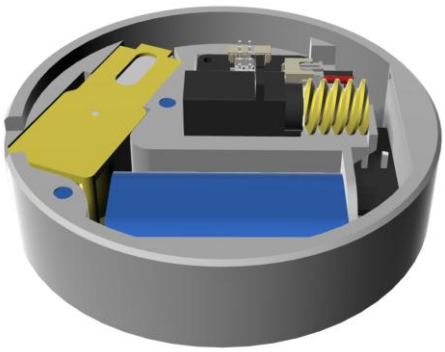


9、将舵机的电源线，插头有金属露出的一面向左，插入绿色框的针座。并按绿色箭头走线，过程中将线材卡入蓝色圆圈内的理线卡槽，过长的线材折叠后塞入右侧红框内的空位收纳。

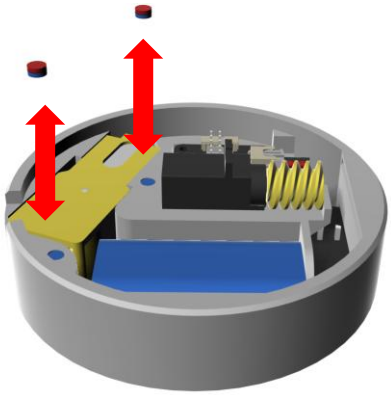


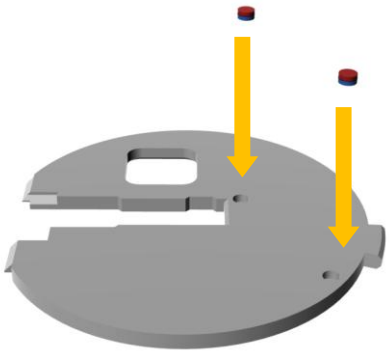
10、将2枚4mm×2mm的磁铁嵌入旋转底座的孔位内，如果感觉旋转底盘放不平，可以先将底盘放在之前安装好的底座上，方便后续安装。



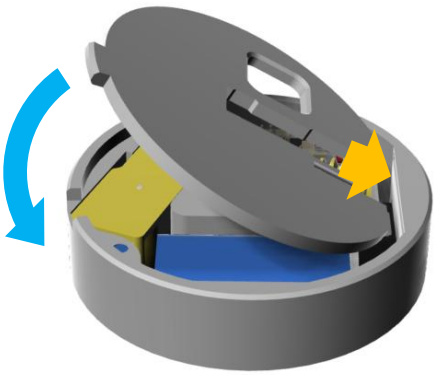


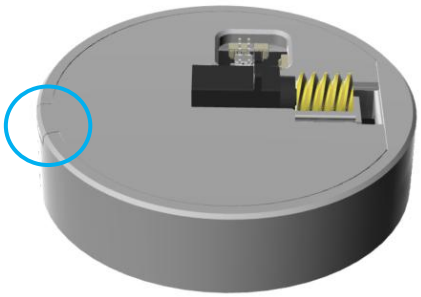
11、取另外2枚磁铁，分别找到他们与之前放置的磁铁相斥的那一面。将该面向下，分别插入图中盖子上表面的两个凹槽孔洞内。



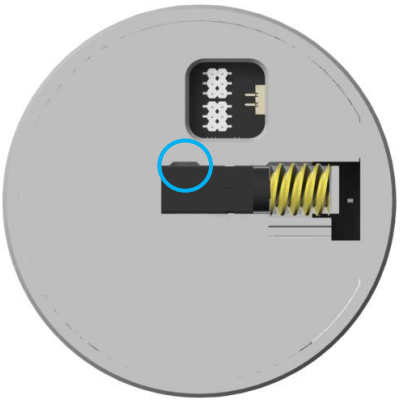


12、先将盖子斜着插入旋转底盘内的槽位内，再向下盖上盖子。测试盖子是否能磁吸在旋转底座上。测试完毕后，请取下盖子方便后续安装。如果要取下盖子，从蓝色圆圈内的缝隙向上扣开即可。

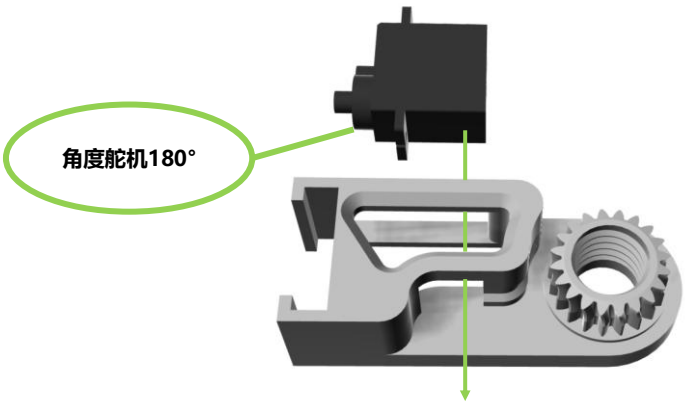




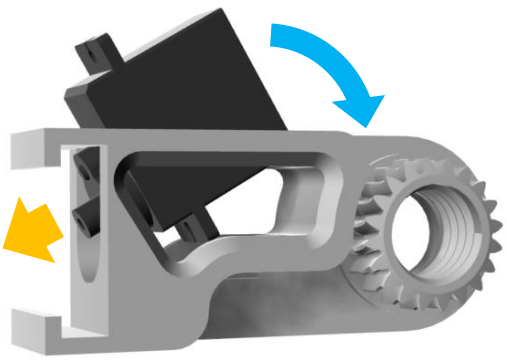
13、盖子如果盖不上，请观察内部的右侧理线槽内的线材是否折叠收纳规整；或是舵机的电源线是否能从盖子的预留的理线槽（蓝色圈圈标识）内穿入。

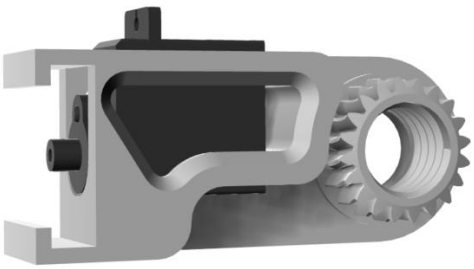


14、将角度舵机的电源线横穿上臂内的槽位，如绿色箭头线所示

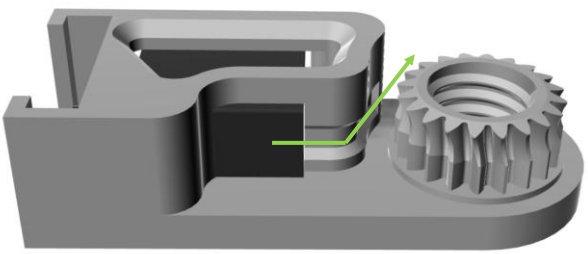


15、将角度舵机凸起先斜着插入槽位前面的空洞内，再旋转着将舵机完全插入槽位内。

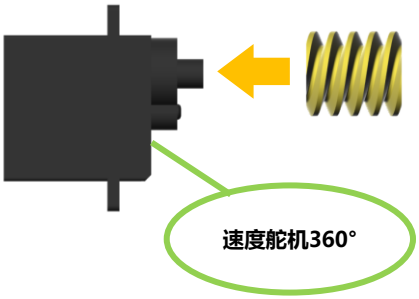




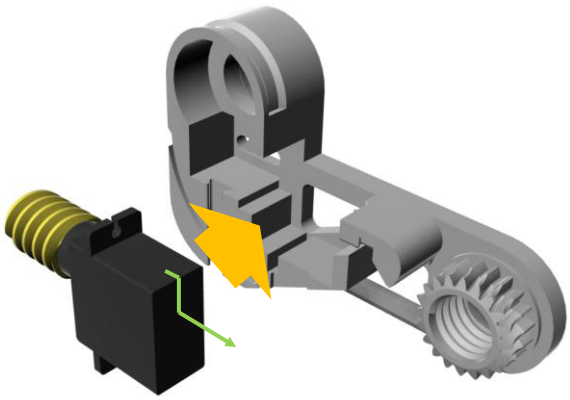
16、将舵机的电源线按绿色箭头线方向。沿着内部的理线槽整理。整理完毕后放在一边备用。

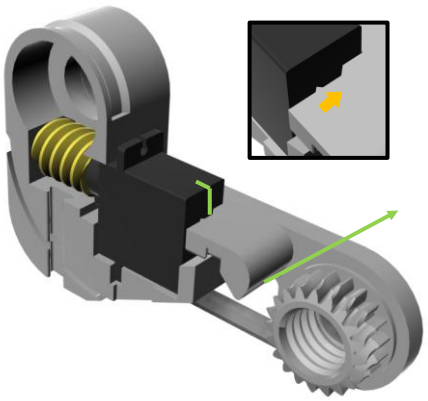


17、取出一个新的速度舵机，将蜗杆插入速度舵机的转轴上，并尽可能的按压。

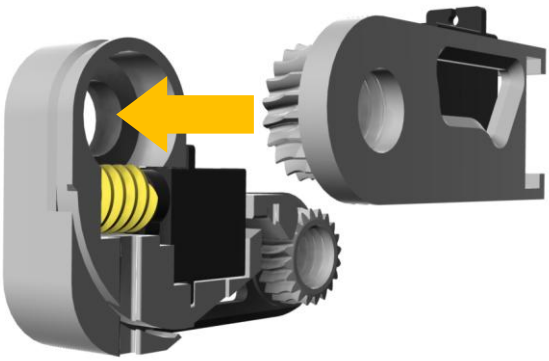


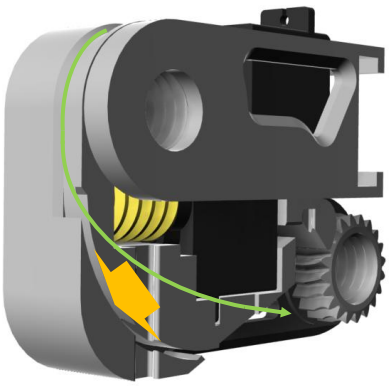
18、将速度舵机的线材如绿色箭头线一样按住，同时将舵机嵌入下臂外壳内的槽位，线材也嵌入预留的理线槽位中的小凸起内部槽位。



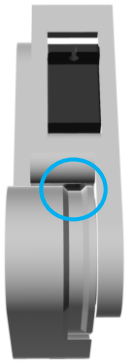


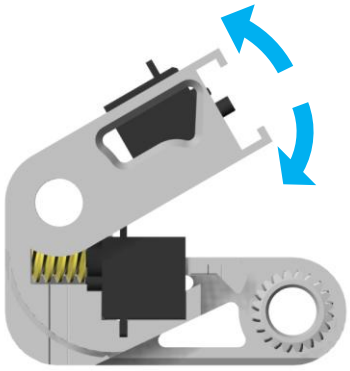
19、将上臂蜗轮嵌入下臂外壳的槽位中。嵌入后，将上臂角度舵机的电源线沿着外面的理线槽环绕理线，并嵌入下臂外壳中，预留的弧形理线槽内。



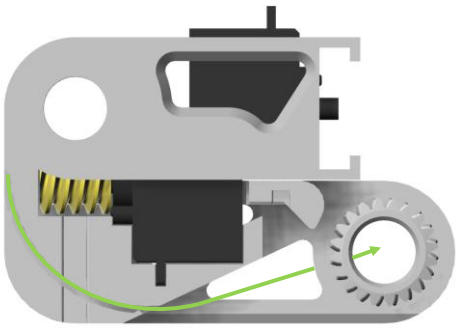


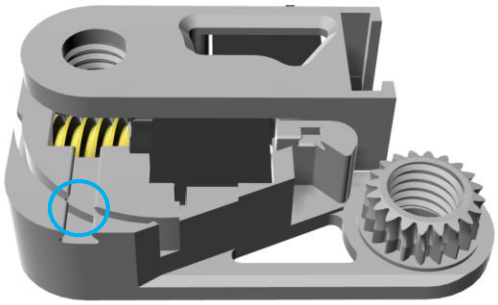
20、如果上臂难以嵌入下臂外壳，可以观察上臂的电源线是否正确嵌入预留的凹槽（蓝色圆圈示意的）。或者尝试将上臂上下旋转一定角度，但是需要让上臂与下臂的夹角尽量的小。



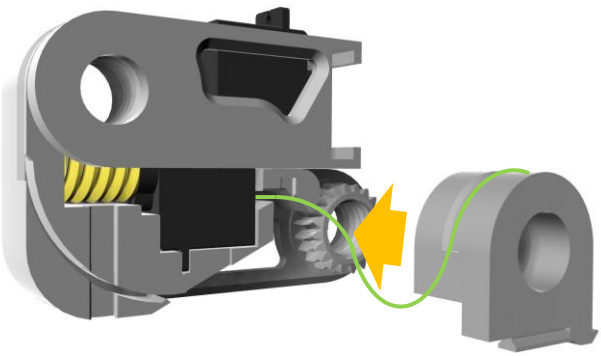


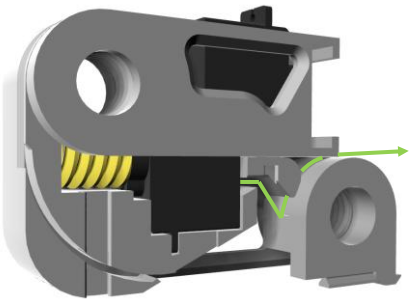
21、将嵌入理线槽内的上臂电源线从下臂的三角形孔内穿过（如绿色箭头线所示）。并保证之前嵌入理线槽内的电源线在蓝色圈圈位置没有突出于凹槽内表面。



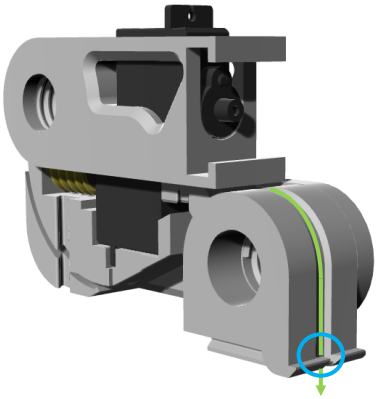


22、将下臂的舵机电源线如绿色线条预缠在关节外壳的槽位上，并同时将关节外壳嵌入下臂外壳的蜗轮内。

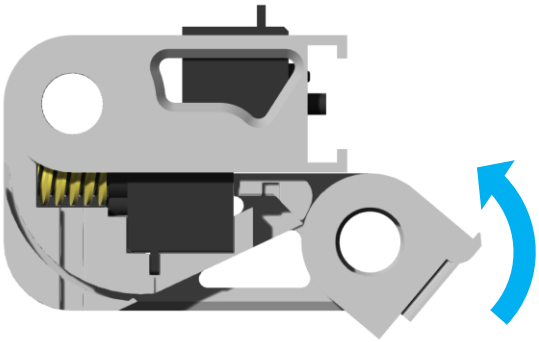


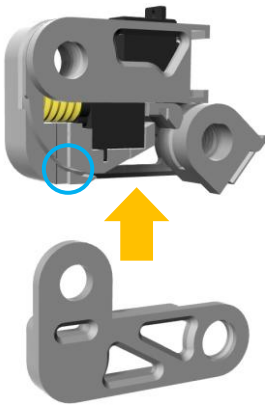


23、将多余的下臂速度舵机的电源线如绿色箭头线一样，缠绕在外壳槽位上，并将线材卡入蓝色圆圈内的卡扣中

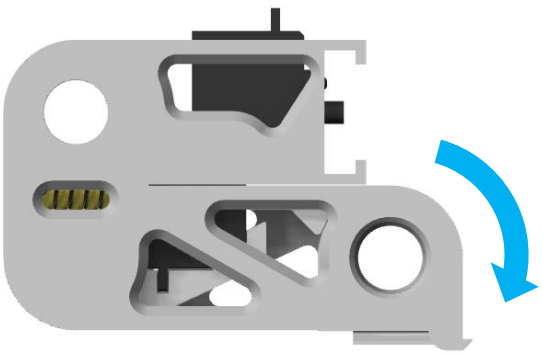


24、将安装好的关节外壳旋转一定角度，并将下臂盖子从滑槽内向上滑入安装，注意蓝色圈圈内的滑槽内，有无线材凸起，否则会影响安装。

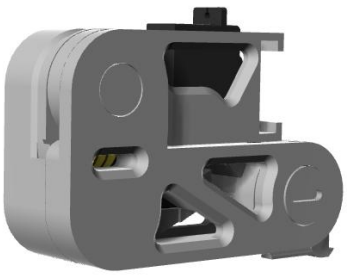




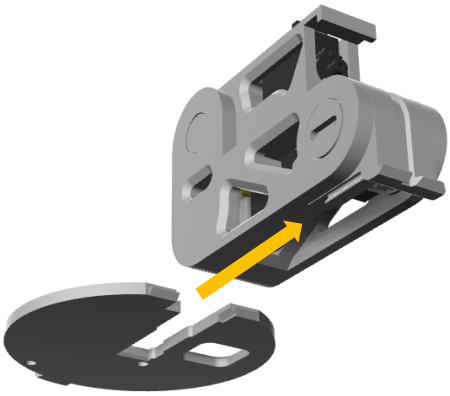
25、安装完毕后，旋转关节外壳，观察是否能够灵活转转动。

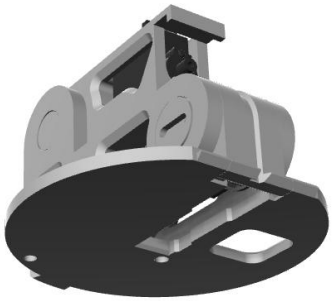


26、将末端没有螺纹的部分较长的螺丝旋入上臂的螺丝孔，另一个旋入下臂的螺丝孔，请不要旋太紧。

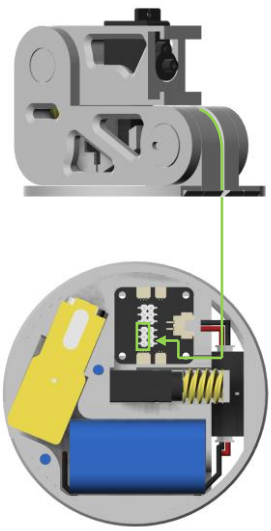


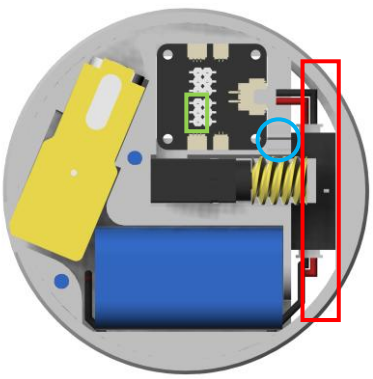
27、将转盘盖子滑入关节外壳下方的槽位中



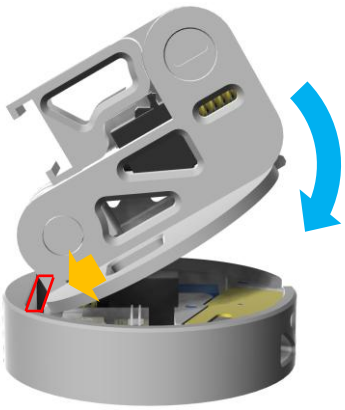


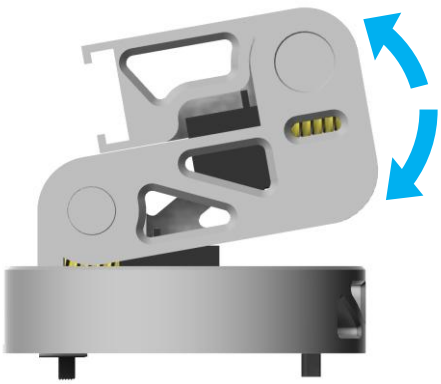
28、将下臂速度舵机的电源线，插头有金属露出的一面向左，插入绿色框内的针座。将线材卡入蓝色圆圈内的卡扣，并将过长的线材折叠收纳在红色框内的空洞内。



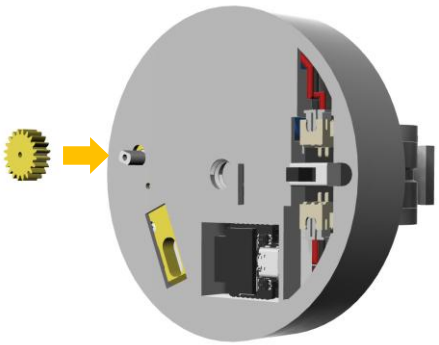


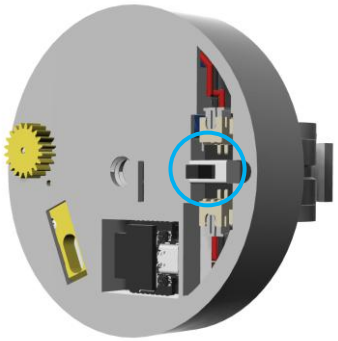
29、边将上一步骤的电源线塞入红色框的空位中，边盖上上盖。有时候需要稍微旋转上部的机械臂，才可以顺利的合上上盖，因为内部的蜗轮蜗杆在特定角度才会啮合。



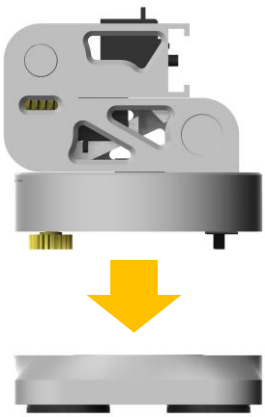


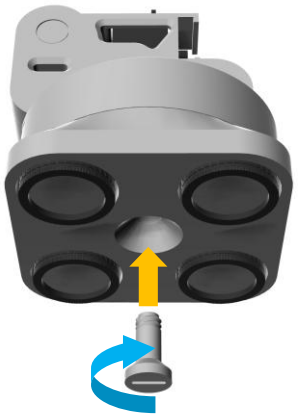
30、将底盘齿轮插入旋转底盘底面的电机转轴中，并用力按压。完成后打开蓝色框的机械臂电源开关。



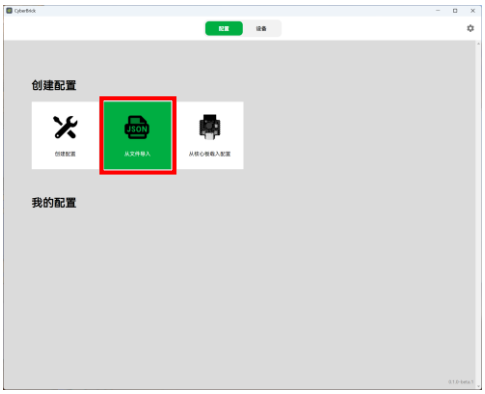


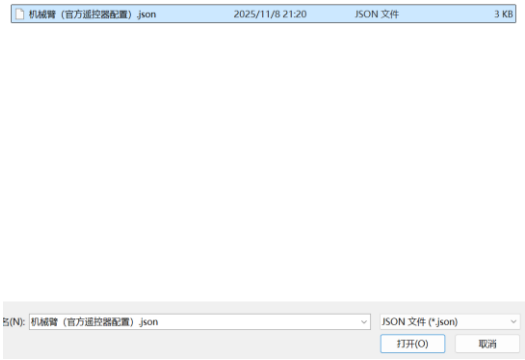
31、将上部结构放置在底座上。如果你使用的不是易拆底座，还需要从下方锁入一颗螺丝。（判断底座类型：观察底座上表面中心，有凸起的是易拆底座，有孔洞的是固定连接底座。）



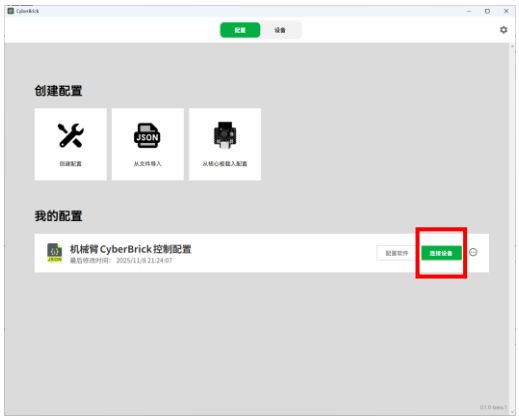


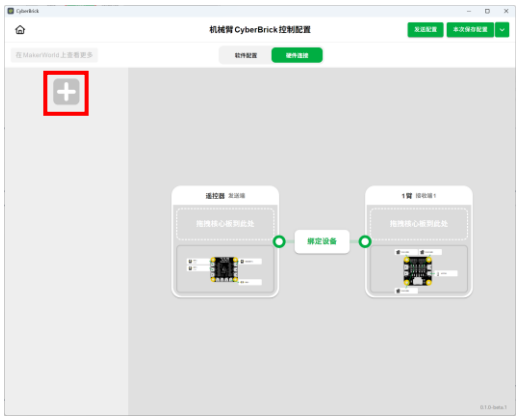
32、取出你的遥控器，并打开遥控器电源开关。之后请打开CyberBrick软件，选择【创建配置】中的【从文件导入】，如果你的遥控器是L-ONE手柄遥控器，则选择【机械臂（手柄遥控器配置）】打开，如果是CyberBrick官方遥控器，则选择【机械臂（官方遥控器配置）】



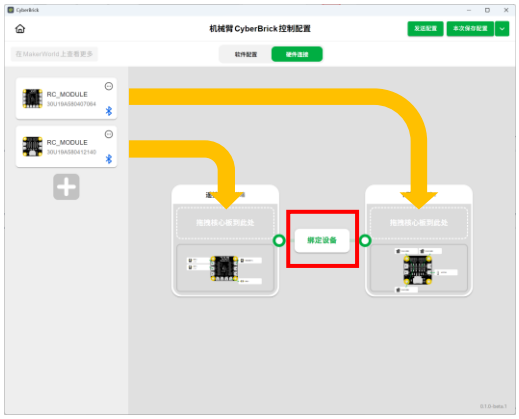


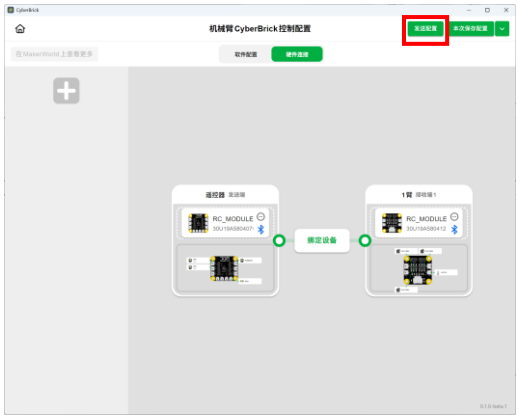
33、选择【我的配置】中的【连接设备】，点击左侧加号，连接你的CyberBrick发送板和接收板



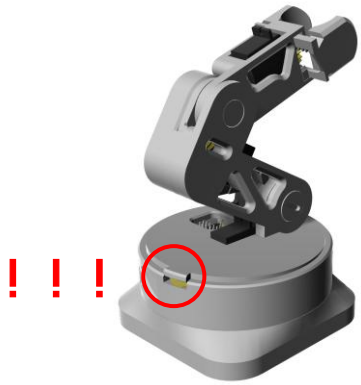


34、将核心板拖入对应的虚线框内，并点击绑定设备，等待两个核心板绑定成功。再点击右上角的发送配置，并等待配置发送成功。



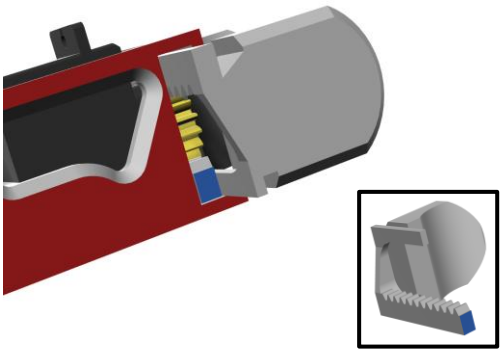


35、向前轻推遥控的摇杆，并尝试遥控器上的各个按钮控制机械臂。在测试过程中，如果旋转底座的上盖弹开，请下压后再继续操控机械臂。熟悉后将机械臂调整到如下姿态。在姿态调整的过程中，注意外部走线的长度是否阻碍了机械臂折叠和伸长，如果有，则按需轻拽外部的绕线调整长度。

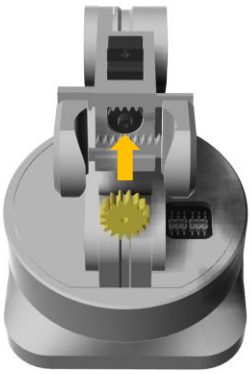


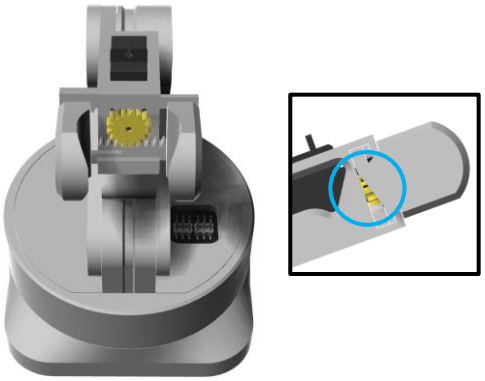
36、将上臂夹爪角度舵机的电源线，插头有金属露出的一面向左，插入绿色框内的针座。按动遥控器上的小按钮，观察机械臂末端的舵机是否在运动。没问题后长按小按钮，发现舵机运动后停止。放下遥控器。

37、将左右两个夹爪从机械臂末端的槽位两侧插入，插入后保证左右两个夹爪齿条末端的平面（蓝色平面），分别与机械臂上臂的左右外表面（红色平面）平齐。安装完成后，夹爪应是张开的状态。

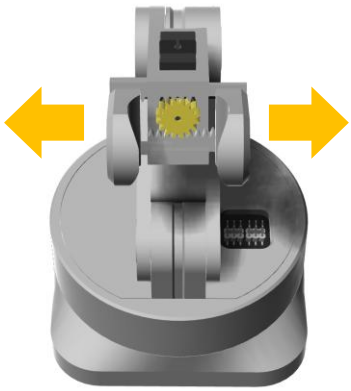


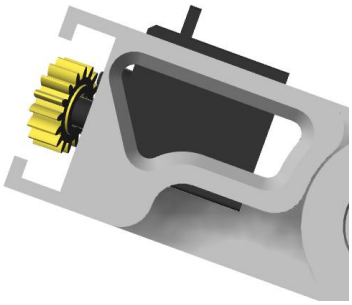
38、将夹爪齿轮插入角度舵机的转轴中，并用力按压，保证其尽量安装到底。可以从蓝色圆圈的缝隙处观察按压进度。短按遥控器上的小按钮数次，观察夹爪是否能每按动一次闭合一点，并且完全闭合后再次短按小按钮，夹爪可以张开。测试在任何时候长按小按钮，夹爪都能张开。



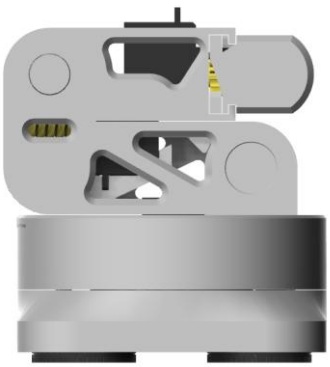


39、如果夹爪出现问题，要取下夹爪，请先断开夹爪舵机的电源，之后向两边抽出左右夹爪，最后使用工具翘出小齿轮。如果要安装，请务必重新从步骤36开始安装！





40、安装完毕



如果您不嫌麻烦，以下几点可以很大地提高机械臂的使用体验。
If you don't mind the extra effort, the following points can significantly enhance your robotic arm experience.

1、如果您发现磁铁比较容易从磁铁槽中脱落，可以适当的使用胶水加固。
If you find that the magnet tends to fall out of the magnet slot easily, you can reinforce it with a suitable adhesive.

2、建议在所有的齿轮传动、蜗轮蜗杆传动的位置，适当使用CyberBrick附赠的润滑油进行润滑。这样可以减少机械臂运行过程中的噪音和震动，一定程度上还可以提高承力极限。
It is recommended to apply the lubricant included with CyberBrick appropriately to all gear and worm gear transmission points. This will reduce noise and vibration during robotic arm operation and may also enhance load-bearing capacity to some extent.

3、尽量保证电池的电量充足。
Ensure the battery is sufficiently charged.

4、定期重新3D打印有磨损齿轮或是蜗轮和蜗杆，并且使用较小的层高打印所有部件
Periodically reprint worn gears or worm gears and worms using 3D printing, and print all components with a smaller layer height.